Zero-shot generalization of EEG attention-state decoding across increasingly stringent validation strategies on open datasets

Sanzhar Sovet[1], Beibit Abdikenov[1*], Manzura Zholdasova[2,3], Altyngul Kamzanova[2], Medet Mukushev[1], Almira Kustubayeva[2,3]

1 Science and Innovation Center "Artificial Intelligence", Astana IT University, Astana, Kazakhstan

2 Department of Biophysics, Biomedicine, and Neuroscience, Al-Farabi Kazakh National University, Almaty, Kazakhstan

3 Brain Institute, Al-Farabi Kazakh National University, Almaty, Kazakhstan

* beibit.abdikenov@astanait.edu.kz

Abstract

Electroencephalography (EEG) decoding models must generalize across people and experimental paradigms, yet non-stationary signals and the uncertain equivalence of labels such as "attention" make this difficult. Evidence on whether nominally aligned attention-related datasets provide useful zero-shot transfer across tasks and datasets remains limited. Here, we evaluated zero-shot generalization of an operationally defined binary contrast between high- and low-attention states in two technically compatible public EEG datasets: a five-participant mental attention-state dataset recorded during a train-simulator task and SAM-40, a 40-participant dataset spanning Stroop, arithmetic, and mirror-image tasks. The datasets were harmonized through a shared 12-channel input space, a fixed preprocessing pipeline, and a common label mapping. Drowsiness was excluded to avoid conflating the contrast with differences in reduce confounding by arousal, visual input, and ocular activity. Five classifiers spanning linear, tree-based, and deep-learning approaches were assessed using group-disjoint within-dataset, cross-subject, cross-task, and cross-dataset validation, without target-specific

adaptation. Mean balanced accuracy across decoders was nearly unchanged when generalizing to unseen participants, differing from the within-dataset baseline by at most two percentage points. Transfer across heterogeneous domains was consistently costlier: cross-task transfer fell on average Compared with target-specific baselines, balanced accuracy fell by an average of 6.4 percentage points below the corresponding target-specific baselines and cross-dataset transfer between SAM-40 task compositions and 13.7 percentage points below them, with a decline in points across datasets; every directed transfer comparison examinedshowed a decline. Absolute accuracy also varied materially across decoders. Thus, shared channels, technically compatible acquisition, and nominally aligned labels were insufficient to preserve target-specific performance under heterogeneous transfer. These leakage-resistant, directed pre-adaptation benchmarks support empirical source selection and provide matched references for evaluating whether transfer learning, domain adaptation, or personalization yields meaningful gains. The findings concern protocol-defined contrasts rather than a unitary neural marker of attention.

Introduction

Electroencephalography (EEG) is a technique for recording brain activity using multichannel scalp electrodes that measure voltage signals reflecting underlying cognitive processes. It is widely used in neuroscience and brain–computer interface (BCI) research because of its non-invasive nature, high temporal resolution, portability, and relatively low cost [1]. At the same time, EEG signals are inherently noisy, non-stationary, and exhibit high inter-subject and inter-session variability [2, 3]. In addition, these challenges are compounded by the scarcity of large, diverse, and well-annotated open EEG datasets [4]. Consequently, developing a robust decoding model with strong generalization capabilities across subjects, sessions, and tasks remains an open and central challenge [2, 5]. One promising approach to address these problems is transfer learning (TL). TL reuses representations already learned on one or more source problems to reduce the amount of labeled data and training effort a related target problem would otherwise require [6]. Its success depends not only on modeling and learning strategies but also on the degree of domain and label compatibility between source and target datasets.

For cognitive constructs such as attention, this compatibility is especially consequential and difficult to guarantee because attention remains an ambiguous concept and is not a unitary construct with a single universally accepted operational definition [4, 7]. Rather, attention is an umbrella term encompassing a set of partially independent processes and functions, such as selecting relevant information, maintaining focus over time, shifting attention between tasks, and allocating cognitive resources. These processes may operate on both external sensory input and internally represented information, including information maintained in working memory [7–9]. The relative contributions of these processes may vary with task demands, sensory modalities, task goals, experimental instructions, assigned conditions, self-reported assessments, and other aspects of experimental design and label definition. Consequently, experimental paradigms described as measuring "attention" may probe non-equivalent configurations of cognitive processes. As a result, different datasets may represent different attention-related cognitive states while assigning them the same nominal labels. In this study, we therefore treat "high-attention" and "low-attention" as operationally defined protocol conditions rather than as an independently validated attention construct, and ask whether decoders discriminate these conditions rather than whether they decode attention itself.

Previous EEG transfer-learning research has been predominantly concerned with cross-subject and cross-session generalization, whereas cross-task, cross-dataset, and cross-paradigm transfer have received comparatively less systematic attention [2, 10, 11]. Other recent work has evaluated cross-subject robustness of deep decoders on multiple motor-imagery benchmark datasets while keeping the task paradigm fixed [12].

Consequently, systematic empirical evidence for source–target suitability across heterogeneous attention-related EEG datasets and tasks remains limited. Source–target suitability is often motivated conceptually or inferred from nominally aligned labels. This leaves unclear which source–target combinations provide useful zero-shot transfer, how much target discriminability is retained relative to target-specific baselines, and whether transferability is direction-dependent. Establishing such pre-adaptation baselines is necessary to justify source selection and to interpret gains or negative transfer from subsequent transfer-learning and domain-adaptation methods [13].

Here, we address this gap by systematically evaluating zero-shot generalization across two technically compatible public EEG datasets spanning multiple tasks. Under a common preprocessing pipeline, we place within-dataset baseline performance, cross-subject generalization, and zero-shot transfer across tasks and datasets on a single ladder of increasingly stringent validation strategies. We characterize how performance behaves along this ladder, comparing directed source–target pairs against target-specific baselines to establish whether transfer is direction-dependent, and we test whether the resulting pattern is reproduced across five decoders ranging from linear and tree-based baselines to deep neural networks. Together, these analyses yield reproducible pre-adaptation benchmarks for evaluating candidate sources and contextualizing both gains and negative transfer produced by subsequent transfer-learning and domain-adaptation methods.

Open-access datasets

EEG data for mental attention state detection (Dataset A)

The dataset described in [14] comprises 34 EEG recordings collected from five participants over the course of seven days, with the fifth participant completing only six sessions. During each session, participants operated a virtual train using the *Microsoft Train Simulator*. Each recording lasted approximately 35 to 55 minutes. The participant cohort consisted of healthy university student volunteers who provided written informed consent after receiving information about the study objectives and procedures; the original study reported that it was conducted in accordance with the ethical guidelines of Mersin University [14].

During every session, participants followed the same predefined route in the virtual train simulator. The route required little active control and kept task demands comparable across recordings. Each session comprised three sequential phases designed to reflect distinct target mental states: *focused*, *unfocused*, and *drowsy*. Participants received standardized instructions from the experiment supervisor for each phase:

- **Focused (0–10 min):** The focused phase required active monitoring of and mental engagement with

the simulator, including sustained attention to the virtual controls and screen events.

- **Unfocused (10–20 min):** During the unfocused phase, participants disengaged from the task, ceased interaction, and were instructed to keep their eyes open and remain awake.
- **Drowsy (20–30 min):** During the drowsy phase, participants were allowed to relax, close their eyes, and enter a drowsy or lightly dozing state, simulating reduced cognitive arousal.

EEG signals were recorded using a modified Emotiv headset, equipped with wet electrodes and supporting wireless Bluetooth transmission. The device recorded data at a sampling rate of 128 Hz and with a voltage resolution of 0.51 μV. The dataset includes 14 EEG channels, labeled according to the 10–20 international system.

SAM-40: dataset of 40-subject EEG recordings (Dataset B)

The **SAM-40** dataset, described in [15], contains EEG recordings from 40 healthy participants (14 females and 26 males) with a mean age of 21.5 years. It was originally The original study obtained informed consent from all participants and approval from the Institutional Ethics Committee of Gauhati University (GUIEC/2019/019, 15 October 2019). It reported voluntary and anonymous participation and compliance with the Declaration of Helsinki [15]. The dataset was collected to investigate short-term stress responses induced by various cognitive tasks. Its protocol corresponded to the *continuous cognitive engagement* paradigm described in [16] and included four conditions:

- **Resting state**: Participants relaxed while listening to music during EEG recording.
- **Stroop Color-Word Test**: Participants identified the color of the ink in which a word was printed while ignoring the word's semantic meaning. The task included two types of stimuli: *congruent* (the color name matched the ink color) and *incongruent* (the color name and ink color differed). Each task run included 11 stimuli.
- **Arithmetic problem solving**: Participants mentally solved a mathematical problem and determined whether the presented answer was correct, responding with a "thumbs up" or "thumbs down" gesture. Each task run included 6 arithmetic stimuli.
- **Mirror Image Recognition**: Participants were shown images and asked to determine whether they were mirror-symmetric or asymmetric, responding with the same gestures. Each task run included 8 images.

Each task was performed for 25 seconds. Between consecutive cognitive tasks, participants rested for 5 seconds and then received 10 seconds of instructions for the next task. After each trial, participants rated the stress experienced during each task on a 10-point scale, where 1 indicated minimal stress and 10 indicated maximal stress. Three trials, each using a different set of stimuli, were recorded for every participant.

EEG signals were recorded using an Emotiv Epoc Flex headset with 32 wet electrodes, labeled according to the international 10–20 system. Data were acquired at a sampling rate of 128 Hz and a voltage resolution of 0.51 μV.

Ethics statement

This study was a secondary analysis of publicly available EEG recordings and involved no additional data collection or participant contact. No additional ethical approval or participant consent was required for this secondary analysis. The consent procedures and ethical oversight reported for the original studies are described in the mental attention-state dataset and SAM-40 sections above.

Methodology

The methodological design of this study was developed to evaluate the generalizability and transferability of EEG-based attention decoding across different recording conditions, participants, and task paradigms. To this end, we adopted conducted a secondary analysis of two publicly available datasets that differ in their experimental protocols but share compatible technical specifications, thereby enabling a controlled comparative analysis. The datasets were harmonized through a binary operational labeling scheme that distinguishes high-attention states, defined by protocol-imposed task demand, from low-attention states of rest or task disengagement, enabling consistent evaluation across scenarios. This section outlines the rationale for dataset alignment, the strategies employed for cross-validation, and the experimental settings designed to assess within-dataset performance as well as cross-subject, cross-task, and cross-dataset generalization.

The datasets were harmonized through a binary operational labeling scheme that distinguishes high-attention states, defined by protocol-imposed task demand, from low-attention states of rest or task disengagement, enabling consistent evaluation across scenarios. These labels denote protocol-defined conditions rather than independently validated levels of attention. They may reflect differences in task demands, arousal, sensory input, stress, or motor responses between resting and active cognitive conditions. The analyses therefore assess whether decoders discriminate the specified protocol conditions, rather than whether they decode attention as a general psychological construct.

To evaluate zero-shot generalization, we used the cross-subject, cross-task, and cross-dataset validation protocols. In each protocol, a decoder was trained on its source composition and applied unchanged to the held-out target data. The within-domain baseline was target-specific and served as a reference for these zero-shot evaluations. Held-out target recordings contributed to only one fitted preprocessing operation, ICA artifact cleaning. ICA was fitted without class labels, separately for each selected session in the mental attention-state dataset and for the selected recordings of each participant in SAM-40. No ICA decomposition pooled source and target recordings or recordings from different participants. Feature extraction was applied in the same form to source and target recordings. All other fitted operations were estimated from source training data only, including standardization statistics, class weights, and decoder parameters. No target data were used for model selection, hyperparameter search, early stopping, checkpoint selection, decoder retraining, or tuning. Target labels were never used. Thus, the zero-shot protocols tested direct application of a source-trained decoder without target-specific decoder adaptation.

All experiments were executed through a version-controlled computational workflow. For every run, the workflow stores the resolved configuration of data preparation, validation, and model settings together with the run outputs needed to reconstruct the reported evaluation metrics. These records provide run-level provenance and allow each reported metric to be traced to the experiment that produced it. The complete implementation, experiment configurations, and run-level artifacts are available at `https://github.com/sovetskiysn/eeg-validation-strategies`, released under the MIT license and archived at `https://doi.org/10.5281/zenodo.22255472`.

Dataset alignment and integration rationale

Effective evaluation of cross-task and cross-dataset generalization relies on the availability of datasets that are compatible both technically and conceptually. In this study, we used two publicly available EEG datasets that differ in their original purpose and labeling schemes, yet retain sufficient technical and conceptual compatibility for a meaningful comparison.

A major strength of this dataset combination is the compatible acquisition setup of the two datasets. Both datasets were recorded using Emotiv EEG systems with the same sampling rate (128 Hz) and voltage resolution, although their original electrode montages differed (14 versus 32 channels). For cross-dataset analyses, only the 12 channels common to both datasets were retained. Differences in EEG recording systems and electrode configurations can hinder cross-dataset transferability [17]. Restricting the analysis to the shared channels substantially reduces this configuration-related mismatch, although it does not eliminate all acquisition-related distribution shifts. Within the limited set of publicly available datasets suitable for this

comparison, this pair offered a comparatively compatible basis for the present comparative evaluation.

The conceptual rationale for combining the datasets rests on the cognitive conditions represented in their respective protocols. The mental attention-state dataset contains explicit attention-state labels (*Focused*, *Unfocused*, and *Drowsy*) collected during a low-interaction simulation task, whereas SAM-40 was originally designed for stress assessment. Both include conditions involving attentional processes, although their original constructs differ. The SAM-40 dataset includes tasks widely recognized in cognitive neuroscience as requiring attentional engagement:

- **Stroop Color-Word Test**: probes both *selective* and *sustained attention*, requiring participants to suppress automatic reading responses and continually focus on the ink color throughout the task [18–20].
- **Mental arithmetic task**: demands *sustained internal attention* and working memory for continuous manipulation of numerical information [21, 22].
- **Mirror Image Recognition**: engages visuospatial processing and requires both *selective* and *sustained attention* to discriminate between symmetric and asymmetric patterns, relying on mental rotation processes that demand continuous cognitive effort [23–25].

The simulator-based and task-based datasets correspond, respectively, to the *naturalistic stimulus-based protocols* and *continuous cognitive engagement* paradigms described in [16]. In the unified labeling scheme, *Rest* and *Unfocused* are treated as low-attention states, whereas *Focused*, *Stroop*, *Arithmetic*, and *Mirror recognition* are treated as high-attention states. The *Drowsy* phase is excluded from this mapping and from all subsequent preprocessing and analyses because permitted eye closure and drowsiness may confound the comparison with differences in arousal, visual input, and ocular activity. This mapping is summarized in Table 1. Throughout this paper, "high-attention" and "low-attention" states denote these operationally defined protocol conditions rather than an independently validated attention construct.

Table 1. Unified binary attention label mapping across datasets.

Attention State	Conditions
High Attention	Focused, Stroop, Arithmetic, Mirror
Low Attention	Unfocused, Rest

Data preparation

The same deliberately simple preparation pipeline was applied to both datasets and across all validation protocols. Its It used common filtering settings, ICA fitting and component-exclusion criteria, channel alignment

channel-alignment selection, and windowing parameters. These settings were fixed in advance and were not tuned separately for a dataset, task composition, validation protocol or decoder. The purpose was to place recordings in a common input space while limiting preprocessing itself as an additional source of variation. Differences between validation results are therefore interpreted against a fixed preparation procedure. The pipeline was instantiated as a fixed set of dataset preparations, that is, namely the analytical dataset compositions used evaluated in every validation protocol and listed in Table 2. The remainder of this subsection describes the selection, cleaningstandardization, alignment cleaning and segmentation steps that produced them.

Table 2. Analytical dataset compositions.

Dataset configuration	Retained conditions	Participants	Recording units	Windows, n (low / high)	Class distribution, % (low / high)
Full Dataset A	Focused, Unfocused	5	24	3,192 / 3,192	50.0 / 50.0
Full Dataset B	Rest, Stroop, Arithmetic, Mirror	40	120	600 / 1,800	25.0 / 75.0
B: Stroop	Rest, Stroop	40	120	600 / 600	50.0 / 50.0
B: Arithmetic	Rest, Arithmetic	40	120	600 / 600	50.0 / 50.0
B: Mirror	Rest, Mirror	40	120	600 / 600	50.0 / 50.0

Recording and label selection. **Recording selection and label harmonization.** Recordings were selected before signal processing. Following the source study, the first two recording sessions for each participant in the mental attention-state dataset were intended for habituation to the simulator and were not included in the analyses [14]. The *Drowsy* condition was excluded from the binary classification analysis because participants were permitted to close their eyes and enter a drowsy state. This condition differs from the *Focused* and *Unfocused* conditions not only in protocol-defined task demand, but also in arousal, visual input and ocular activity. Including it in the low-attention class would therefore confound the intended contrast. The drowsy segment was removed before filtering and ICA, so no drowsy epochs were generated or presented to a classifier. In contrast, the complete collection of SAM-40 recordings was retained for analysis. SAM-40 provides the task recordings of each trial as separate files. Within each scenario, the task recordings selected from one trial were grouped into a single recording unit, whereas task recordings not selected by the scenario were not carried forward to later preparation steps. Accordingly, the binary mapping in Table 1 was applied to the retained recordings from the two datasets.

Signal standardization and channel alignment. Before filtering and ICA cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Both source datasets were originally sampled at 128 Hz and used the same documented conversion factor of 0.51 μV per digit. Therefore, no resampling or additional amplitude scaling was required for cross-dataset harmonization.

Filtering and ICA cleaning. Filtering was applied to each recording, and ICA was fitted within participant-specific acquisition groupings. For the mental attention-state dataset, each grouping corresponded to a selected session, whereas for SAM-40 it comprised the selected task recordings of a participant. Training and test data therefore never shared a decomposition, except in SAM-40 baseline validation, where different runs of the same participant could fall into training and test folds, so this baseline may be favored in comparisons with participant-disjoint protocols. To prevent preprocessing from becoming an additional source of variation, we used a deliberately restrained, conventional procedure: standard filtering followed by ICA-based artifact cleaning. Keeping preparation deliberately simple allows differences in performance to be interpreted primarily in relation to the recordings and validation protocols, rather than to a complex preparation pipeline. EEG channels were notch-filtered at 50 Hz, band-pass filtered from 1 to 45 Hz and resampled to 128 Hz. Before ICA, they were rereferenced to the average reference. Artifact correction used independent component analysis with extended-Infomax optimization, retaining components that explained 99% of the variance. The resulting components were automatically classified from their spatial, spectral and temporal characteristics. Components identified as eye-blink or muscle artifacts with a probability of at least 0.80 were removed. The ICA random state was fixed by the experiment-wide configuration.

Channel alignment and window segmentation. Window segmentation and metadata. After signal ICA cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Each retained labeled block was segmented into 5-s windows with 0.5-s overlap. Every retained window inherited the label of its source blockand metadata . Metadata identifying its dataset, participant, session, task, run, recording unit and temporal onset were retained with each window. The recording-unit identifier was These metadata were used by the validation protocols to ensure support protocol-specific grouping and split definitions. In particular, the recording-unit identifier ensured that all overlapping windows from a unit remained on the same side of a baseline split.

The fixed selection, cleaning, alignment and segmentation steps therefore yielded a deterministic analytical window set for each dataset composition. All model-specific input representations were derived from these same prepared windows, rather than from separately prepared data. The retained recording units, window counts and observed class distributions of the resulting compositions are those reported in Table 2. Class weights were calculated independently within each training fold. All Dataset B configurations contain the same 40 participants and differ only in the task recordings retained alongside the resting-state recordings. The fixed selection, standardization, cleaning and segmentation steps therefore yielded a deterministic analytical window set for each dataset composition. All model-specific input representations were derived from these same prepared windows, rather than from separately prepared data. The retained recording units, window counts and observed class distributions of the resulting compositions are those reported in Table 2. Class weights were calculated

independently within each training fold. All SAM-40 compositions contain the same 40 participants and differ only in the task recordings retained alongside the resting-state recordings.

Validation strategies

We evaluated the decoders under four validation protocols, each addressing a distinct generalization question. To preserve The within-domain baseline held out recording units from the independence of the training same dataset and test datacondition composition; cross-subject validation changed participant identity; cross-task validation changed both participant identity and SAM-40 task composition; and cross-dataset validation changed datasets, all participant populations, and condition definitions. The participant, session, task, run, and recording-unit metadata retained with each window defined the required groups. All windows originating from the same a recording unit were assigned to remained in the same partition. This prevents , preventing overlapping windows from being divided between training and test sets, reducing crossing the risk of overly optimistic performance estimates due to temporal autocorrelation between neighboring EEG windowstraining–test boundary [16]. The within-domain Because baseline estimates performance when training and folds could include other recordings of test data share participants, comparisons with the same dataset and condition composition but originate from different recording unitsbaseline also include a shift to unseen participants. The remaining protocols examine whether discriminative performance is retained when training Cross-task and testing are separated by participant, cross-dataset differences therefore cannot be attributed to task composition , or dataset alone. Each protocol was used only where the structure of the relevant dataset supported its interpretation. For example, cross-task validation was possible only for SAM-40 because it included multiple task compositions. Model fitting was confined to the training data, The protocol-specific splits and the resulting decoder was evaluated on held-out data without target-specific adaptationICA fits are detailed below.Any class imbalance was addressed through inverse-frequency class weighting calculated independently within each training fold, while test data retained their observed class distribution.

Baseline evaluationvalidation. Reference performance Performance was estimated using five-fold assessed on held-out recording units within the same dataset and condition composition. Selected recordings were divided into five stratifiedgroup cross-validation, group-disjoint folds, separately for the mental attention-state dataset, the SAM-40 dataset, and each task-specific composition of SAM-40. The grouping variable was In the physical recording unit: a selected session in the mental attention-state datasetor a run-level , each selected participant–session recording unit in SAM-40was one group. Each In SAM-40unit comprised , each participant–run was one group containing the resting-state recording and all task recordings data included in the analysis selected composition for that run. Training For example, when evaluating the Stroop composition, the resting-state and test folds thus shared Stroop recordings from the dataset same participant and condition composition run were always assigned to the same fold. In each fold, approximately four fifths of the recording units were used for training and could contain data the remaining

units for testing. All windows from a recording unit were assigned to the same participantspartition. A participant could contribute different sessions or runs to both training and testing, but never from the same physical recording unitcould never occur in both partitions. Training and testing therefore shared the dataset and condition composition while differing in the recording units presented to the decoder. For data-dependent preprocessing, each selected session in the mental attention-state dataset was cleaned using an ICA decomposition fitted separately for that session, so the decomposition remained on one side of the split. In SAM-40, the decomposition used for cleaning was fitted across each participant's retained runs within the evaluated composition and could therefore span training and test runs from that participant. Keeping each unit intact prevented temporally adjacent or partially overlapping windows from the same acquisition from being assigned to opposite folds, thereby avoiding window-level temporal leakage. These estimates serve as within-domain comparators for the transfer protocols.

Cross-subject validation. In each leave-one-subject-out fold, the decoder was trained on all selected data from the remaining participants and tested on all selected data from the held-out participant. Data-dependent preprocessing of the held-out participant's recordings was limited to ICA fitted on that participant's own unlabeled data, separately for each selected session in the mental attention-state dataset and for the selected recordings of each participant in SAM-40. No decomposition used to clean source data included the held-out participant. The resulting performance quantifies zero-shot generalization to an unseen participant.

Cross-task validation. In SAM-40, each task composition represented the same binary contrast of low-attention rest versus high-attention task conditions, rather than classification of task identity. Cross-task validation assessed directed transfer between different compositions of these conditions. The analysis included transfers between the Stroop, Arithmetic, and Mirror task-specific compositions, as well as transfers in both directions between each task-specific composition and the full SAM-40 composition (Full B), which contains all three active tasks. All compositions contained the same resting-state condition. For a transfer between two task-specific compositions, the active task in the target composition was absent from the source composition. In transfers involving Full B, the source and target compositions overlapped in at least one active task: Full B included the task-specific target when used as a source, and a task-specific source was included in Full B when it was the target. These comparisons therefore asked whether test a decoder fitted change in the mixture of task conditions, not generalization to this contrast under one an entirely unseen taskcomposition retained discriminative performance under another.

The cross-task folds were also subject-disjoint. For each fold, the source composition from all but one participant formed the training set, whereas the target composition from the held-out participant formed the test set. The protocol thus evaluates Data-dependent preprocessing was performed separately within the combined shift

to an unseen selected recordings of each participant and composition. Thus, no ICA decomposition pooled source and target recordings, and the held-out target participant contributed only to the ICA fitted on that participant's own unlabeled target recordings. Every comparison therefore combines a different change in participant identity with a change in task composition, rather than within-participant transfer . Only the transfers between tasks two task-specific compositions additionally evaluate an active task absent from source training; none isolates the effect of task identity from the participant shift. This distinction Keeping participants disjoint prevents participant-specific EEG patterns from contributing to apparent cross-task generalization transfer performance [26, 27].

Cross-dataset validation. To evaluate transfer between datasets, models were trained on all available data from a source composition and tested on a composition from the other dataset, in both transfer directions and without target-specific adaptation. Data-dependent preprocessing was performed separately for the two datasets: the mental attention-state dataset was cleaned using ICA fitted separately to each selected session, whereas SAM-40 was cleaned using ICA fitted separately to each participant's selected recordings. Consequently, source and target datasets were never pooled for ICA, and target recordings contributed only to a decomposition fitted on their own unlabeled acquisition grouping. SAM-40 was evaluated both as a complete composition and as task-specific compositions. Unlike the preceding protocols, cross-dataset Cross-dataset validation changes several factors simultaneously, including the participant population, experimental paradigm, recording setup, and condition definitions. It therefore measures transfer under their compound distribution shift rather than attributing a performance change to any single factor [17]. The common sampling rate, harmonized binary labels, and shared 12-channel montage establish a compatible input and prediction space, but do not remove the substantive differences between the two datasets.

Models

We evaluated five classifiers rather than relying on a single decoder to test whether the conclusions are robust to a change in modeling assumptions. This makes the decoder an explicit factor of the design: a difference between validation protocols that is reproduced across models is more credibly attributed to the data and evaluation protocol than to the idiosyncrasies of one learner. The selected models span three families used in EEG decoding: linear, tree-based, and deep-learning architectures. The deep networks are included because they learn task-relevant representations directly from the channel-by-time EEG windows, reducing reliance on a manually specified feature set. All models were trained to predict whether a window belonged to a high-attention or a low-attention state in the sense of Table 1.

Hyperparameters were fixed *a priori* and held constant across datasets, task compositions, validation protocols and runs. No hyperparameter search, early stopping or checkpoint selection was performed. Values

followed library defaults or published reference recipes, except for the departures specified below. A fixed random seed governed fold construction, model initialization and network training. The resolved configuration of each run was stored with its outputs.

Traditional machine learning. Both classical classifiers used the same fixed, label-free feature representation. For each of the 12 channels, we computed log relative power in the 1–4 Hz, 4–8 Hz, 8–13 Hz and 13–45 Hz bands, together with sample entropy, mean, variance, skewness and kurtosis. Relative band power was normalized by the channel total before log transformation. The nine features from each channel formed one 108-dimensional feature vector per window. This feature representation was applied unchanged to every dataset, task composition, and validation protocol.

Logistic regression provides a transparent linear baseline. It estimates a linear decision boundary from the feature vectors, allowing the contribution of individual features to be inspected. Its feature vectors were z-score standardized using the mean and standard deviation estimated from the training fold only.

XGBoost [28] represents tree-based ensemble learning, allowing nonlinear relations and feature interactions to be modeled within the same feature representation. It was also the decoder used in the originally submitted version of this study, preserving a direct point of comparison with those results. It received the same feature vectors without scaling. For both classical models, inverse-frequency class weights were calculated from the training fold only.

Deep learning. The remaining three classifiers operate directly on the prepared channel-by-time EEG windows, rather than on the classical feature vectors. Each input window had a 12×640 channel-by-time shape and was standardized using statistics estimated from the training fold only.

ShallowFBCSPNet [29] is a shallow convolutional neural network designed to emulate the main operations of the filter-bank common spatial pattern (FBCSP) pipeline within a trainable model. It first applies temporal convolution and spatial filtering across channels, then uses a squaring nonlinearity, mean pooling and a logarithmic activation to form band-power-like features. Thus, its inductive bias reflects a conventional EEG decoding approach, while its filters are optimized jointly with the classifier.

EEGNet [30] is a compact convolutional neural network designed specifically for EEG. Its temporal convolutions learn filters over time, depthwise convolutions learn spatial filters across electrodes for each temporal feature map, and separable convolutions combine these learned features efficiently. This factorized design substantially limits the number of trainable parameters while retaining separate temporal and spatial feature learning.

EEG Conformer [31] combines a convolutional front end with a Transformer encoder. The convolutional module forms local temporal and spatial features and converts them into a sequence of embeddings.

Self-attention in the Transformer then models global dependencies between these embeddings. It therefore complements the local receptive fields of convolution with a mechanism for relating EEG patterns over longer time spans. To limit the risk of overfitting in the smallest group-disjoint training folds, we reduced the capacity of the Braindecode default configuration before evaluation and then fixed it across all scenarios. Specifically, we reduced the number of temporal convolution filters, which also determines the embedding width, from 40 to 16, the number of Transformer encoder layers from 6 to 2, and the number of attention heads from 10 to 4. This retained the convolution–attention architecture while reducing its parameter count for the 12×640 input windows from 527,746 to 170,466.

All three networks were trained from scratch, with no pretrained weights, for 100 epochs with a batch size of 64. Their loss functions used inverse-frequency class weights calculated from the training fold only, and no resampling was applied.

Results and analysis

To characterize zero-shot generalization across the prespecified validation strategies, we evaluated the binary contrast between high-attention and low-attention states with five decoders. Table 3 reports decoder-specific performance across these strategies. It presents participant-level balanced accuracy, macro F1, MCC, ROC-AUC, and per-class recall for logistic regression, the interpretable reference decoder. Corresponding tables for XGBoost, EEGNet, ShallowFBCSPNet, and EEGConformer are provided in S1 Table, S2 Table, S3 Table and S4 Table.

Table 3. Performance metrics across experimental scenarios for Logistic Regression.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.63	0.30	0.73	0.62/0.66
	Stratified K-fold(Full B)	0.68	0.64	0.34	0.74	0.65/0.71
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.74	0.67/0.71
	Stratified K-fold(B: Arithmetic)	0.66	0.65	0.35	0.74	0.66/0.67
	Stratified K-fold(B: Mirror)	0.69	0.67	0.39	0.75	0.66/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.63	0.61	0.29	0.73	0.63/0.63
	Leave-one-subject-out(Full B)	0.66	0.61	0.31	0.74	0.63/0.69
Cross-task (Dataset B)	Full B → Stroop	0.66	0.64	0.34	0.73	0.64/0.68
	Full B → Arithmetic	0.64	0.62	0.31	0.72	0.64/0.65
	Full B → Mirror	0.67	0.65	0.36	0.77	0.63/0.71
	Stroop → Full B	0.63	0.58	0.25	0.67	0.63/0.63
	Stroop → Arithmetic	0.58	0.55	0.16	0.60	0.65/0.50
	Stroop → Mirror	0.63	0.60	0.27	0.68	0.64/0.62
	Arithmetic → Full B	0.62	0.55	0.23	0.68	0.66/0.58
	Arithmetic → Stroop	0.56	0.53	0.14	0.61	0.67/0.46
	Arithmetic → Mirror	0.66	0.64	0.35	0.73	0.67/0.66
	Mirror → Full B	0.64	0.57	0.26	0.70	0.68/0.60
	Mirror → Stroop	0.60	0.57	0.21	0.65	0.66/0.54
	Mirror → Arithmetic	0.62	0.59	0.27	0.70	0.63/0.62
Cross-dataset	Full A → Full B	0.52	0.37	0.03	0.54	0.78/0.26
	Full B → Full A	0.55	0.53	0.12	0.58	0.35/0.76
	Full A → Stroop	0.49	0.42	-0.03	0.51	0.71/0.27
	Stroop → Full A	0.54	0.53	0.08	0.56	0.45/0.63
	Full A → Arithmetic	0.53	0.48	0.08	0.58	0.68/0.38
	Arithmetic → Full A	0.52	0.52	0.04	0.54	0.43/0.61
	Full A → Mirror	0.53	0.47	0.06	0.55	0.72/0.34
	Mirror → Full A	0.52	0.48	0.06	0.54	0.26/0.79

Generalization under increasingly stringent validation strategies

To examine generalization under increasingly stringent validation strategies, we compared directed cross-task and cross-dataset transfer with the baseline scenario for each target composition. These comparisons place performance under progressively less similar training and test data in relation to a target-specific reference. They therefore assess whether the binary contrast was retained without target-specific adaptation as the participant, task composition, or dataset differed between training and test data.

Fig 1 summarizes the directed cross-task transfers between SAM-40 task compositions and cross-dataset comparisons. Rows are grouped by source composition and decoder, and columns denote target compositions. Each nonmissing matrix cell reports balanced accuracy calculated separately from the held-out windows of each target participant and then averaged across target participants for the given decoder and direction. Off-diagonal cells represent zero-shot transfers without target-specific adaptation, whereas the gray diagonal

reports the target-specific baseline estimated with five-fold stratified group validation that keeps recording units disjoint between training and test folds. For each target represented in a summary columnthe summary columns, we first averaged participant-level estimates across the five decoders for each source composition and target. We then averaged these decoder-level transfer and baseline values across targets and calculated their difference. Negative values therefore indicate lower transfer performance than the applicable target baselines, whereas dashes identify comparisons that are not applicable.

Fig 2 compares baseline and leave-one-subject-out cross-subject validation for the mental attention-state dataset and the SAM-40 dataset. For each decoder and scenario, balanced accuracy was calculated separately from the held-out windows of each target participant and then averaged across target participants, and the colored lines show these decoder-specific means. Black points show the mean of the five decoder-specific estimates. Their bars show paired 95% percentile bootstrap intervals from for this mean within each scenario, obtained by resampling target participants with replacement 10,000 target-participant resamples, with the same resample used for all decoders to preserve their pairing. times. Each draw uses the same participant indices for all five decoders, with five participants in the mental attention-state dataset and 40 in SAM-40. The intervals describe variation across sampled participants conditional on the fitted models, without refitting them or treating windows or cross-validation folds as independent observations.

		Cross-task				Cross-dataset					Baseline vs Cross-task mean diff	Baseline vs Cross-dataset mean diff
		Stroop	Arithmetic	Mirror	Full B	Stroop	Arithmetic	Mirror	Full B	Full A		
Stroop	Logistic reg	69.3	57.7	62.7	63.1	69.3	—	—	—	54.0	-9.6	-13.1
	XGBoost	69.0	55.7	61.9	60.8	69.0	—	—	—	50.6		
	EEGNet	66.1	53.8	58.8	59.4	66.1	—	—	—	50.3		
	ShallowNet	78.5	59.3	64.1	65.4	78.5	—	—	—	55.5		
	EEGConformer	73.2	55.6	61.2	61.8	73.2	—	—	—	52.2		
Arithmetic	Logistic reg	56.0	66.4	66.2	62.2	—	66.4	—	—	52.2	-6.3	-10.4
	XGBoost	58.2	69.5	66.6	63.6	—	69.5	—	—	58.9		
	EEGNet	58.9	65.3	64.3	61.9	—	65.3	—	—	50.3		
	ShallowNet	62.9	76.2	71.0	68.1	—	76.2	—	—	55.6		
	EEGConformer	61.8	71.2	69.4	67.2	—	71.2	—	—	59.5		
Mirror	Logistic reg	59.7	62.3	68.6	63.8	—	—	68.6	—	52.4	-6.6	-13.1
	XGBoost	61.5	64.3	70.5	64.9	—	—	70.5	—	53.2		
	EEGNet	61.2	63.3	67.0	64.1	—	—	67.0	—	48.8		
	ShallowNet	63.2	67.4	73.9	67.5	—	—	73.9	—	54.9		
	EEGConformer	60.9	64.6	69.1	64.6	—	—	69.1	—	53.6		
Full B	Logistic reg	66.2	64.5	67.1	68.3	—	—	—	68.3	55.2	-3.1	-11.0
	XGBoost	64.1	65.1	67.2	68.0	—	—	—	68.0	54.1		
	EEGNet	65.8	64.6	68.4	66.9	—	—	—	66.9	51.0		
	ShallowNet	66.9	67.7	67.0	71.9	—	—	—	71.9	54.8		
	EEGConformer	70.7	70.2	72.6	71.8	—	—	—	71.8	58.2		
Full A	Logistic reg	—	—	—	—	48.8	53.4	53.2	52.1	64.0	—	-20.9
	XGBoost	—	—	—	—	47.8	54.0	51.4	51.6	67.0		
	EEGNet	—	—	—	—	44.7	49.2	45.7	46.7	66.6		
	ShallowNet	—	—	—	—	47.5	51.7	47.7	49.2	66.4		
	EEGConformer	—	—	—	—	42.2	52.5	47.3	47.0	64.3		

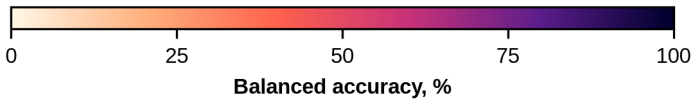


Fig 1. Zero-shot transfer performance across source and target compositions. Rows denote a decoder and the composition it was trained on, and columns denote the composition it was tested on. The gray diagonal is not a transfer but the baseline validation scenario, where a decoder was trained and tested on the same composition. The two right-hand columns show mean differences from the relevant target baselines. Negative values indicate lower performance, and dashes mark contrasts that are not applicable.

For cross-task transfer, the mean decreases from the corresponding target-specific baselines, averaged across all five decoders, were 9.6, 6.3, 6.6, and 3.1 percentage points for models trained on the Stroop, Arithmetic, and Mirror task compositions and the SAM-40 dataset, respectively. The mean decrease across these targets source compositions was 6.4 percentage points; this summary includes both task-specific pairs and

comparisons involving Full B, whose active tasks overlap with those of the task-specific compositions. For cross-dataset transfer, the decoder-averaged decreases for models trained on the Stroop, Arithmetic, and Mirror task compositions, the SAM-40 dataset, and the mental attention-state dataset were 13.1, 10.4, 13.1, 11.0, and 20.9 percentage points, respectively. The mean decrease was 13.7 percentage points. Thus, zero-shot transfer yielded lower performance in every eligible comparison, with a larger reduction when training and test data came from different datasets.

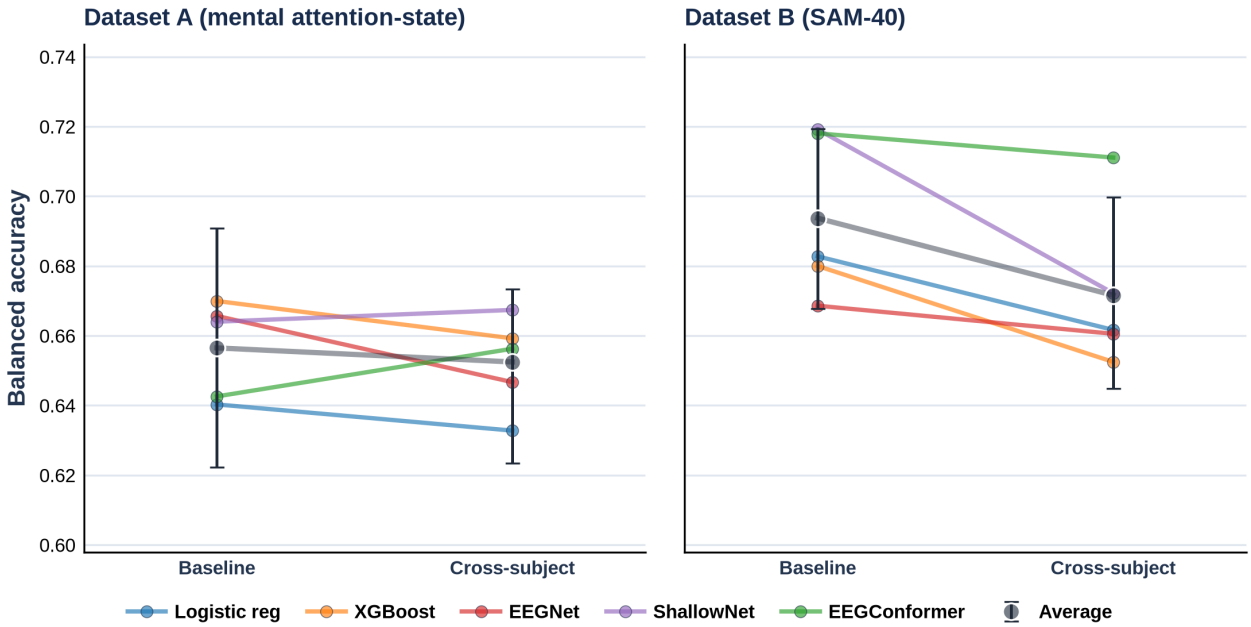


Fig 2. Baseline and cross-subject performance across decoders.

In the mental attention-state dataset, mean balanced accuracy was 65.5% in the baseline scenario and 65.4% in cross-subject validation, a decrease of 0.1 percentage points. In the SAM-40 dataset, it decreased from 69.1% to 67.1%, a difference of 2.0 percentage points. These descriptive differences do not establish equivalence between baseline and cross-subject performance.

With two exceptions, all decoder-specific estimates fell within the corresponding 95% bootstrap interval. In the SAM-40 baseline scenario, ShallowFBCSPNet reached 72.0%, which was slightly above the interval upper bound of 71.6%. In cross-subject validation, EEGConformer reached 70.7%, which was above the upper bound of 69.7%.

Performance consistency across decoders

To assess whether performance patterns across validation strategies depended on decoder choice, Fig 3 compares all validation scenarios across the five decoders. Each colored point represents the balanced

accuracy of one decoder, averaged across target participants in a given scenario. The black point and horizontal bar show the mean across the five decoders and its paired 95% percentile bootstrap confidence interval (CI), respectively, estimated from 10,000 target-participant resamples. The same resample was used for all decoders to preserve their pairing.

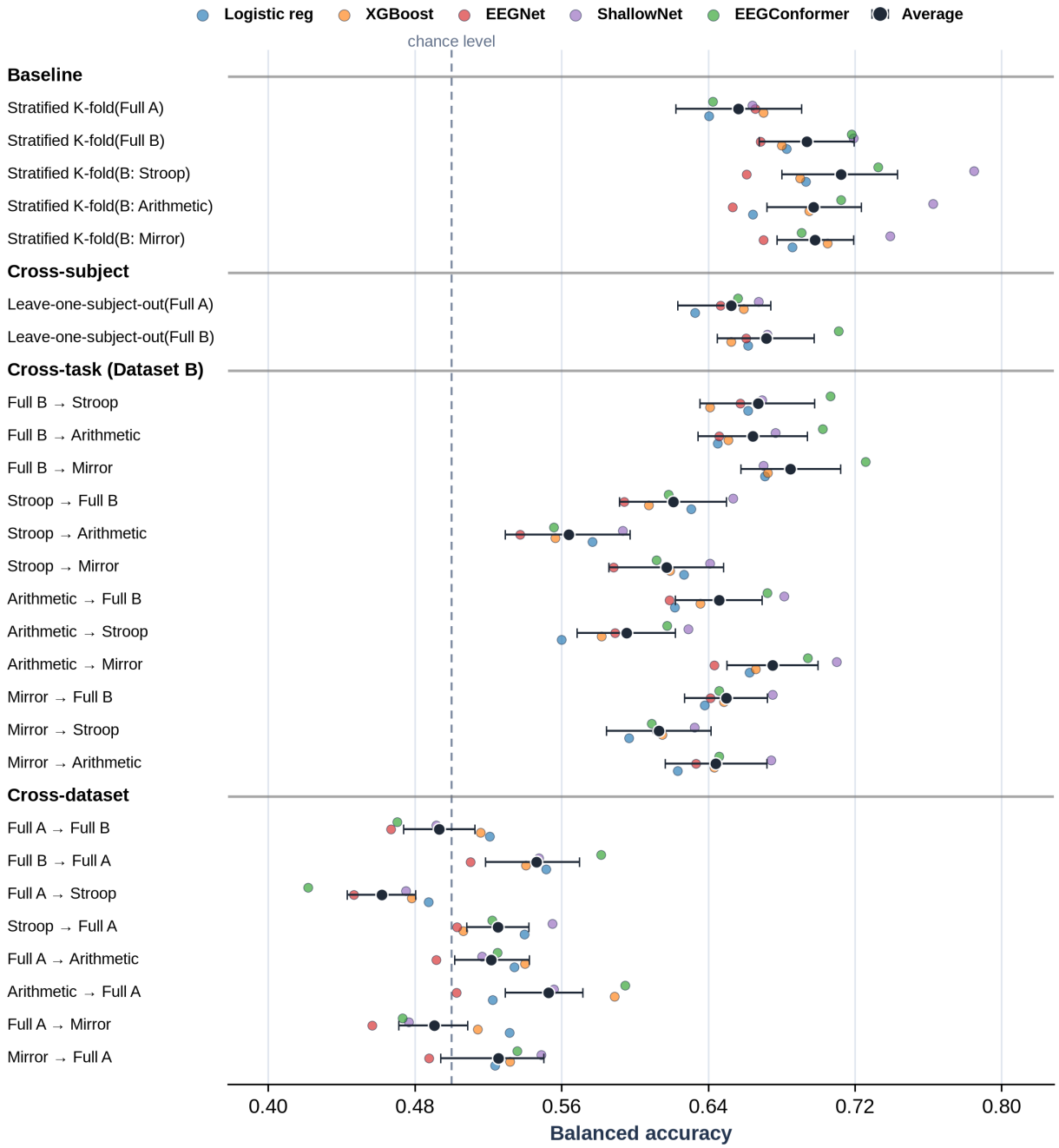


Fig 3. Participant-level balanced accuracy across experimental scenarios for the five decoders.

Across the 27 scenarios, 42 of 135 decoder-specific estimates lay outside the corresponding 95% bootstrap interval. Logistic Regression lay outside the interval in seven scenarios, with four points below and three above the interval. XGBoost lay outside the interval in four scenarios, with one point below and three above the interval. All of these departures occurred in cross-dataset transfer settings.

EEGNet lay outside the interval in 12 scenarios, always below it. ShallowFBCSPNet lay outside the interval in 10 scenarios, always above it. These were the two decoders with the most frequent departures from the scenario-average interval . interval for the mean across decoders. Such departures are descriptive and do not constitute pairwise tests of decoder performance.

EEGConformer lay outside the interval in nine scenarios, with two points below and seven above the interval. Overall, these patterns show that decoder choice materially influenced balanced accuracy across validation scenarios, including both within-dataset baselines and transfer settings.

Discussion

Across the data compositions and validation protocols examined here, zero-shot discriminative performance declined as the difference between the training and test data increased. Performance was comparatively stable when generalization was limited to an unseen participant within a dataset , but was lower Mean performance decreased by 0.1 and 2.0 percentage points for unseen participants in the mental attention-state dataset and SAM-40, respectively, without establishing equivalence to baseline performance. Reductions relative to the target-specific baselines were larger when the training and test compositions also differed by task and lower larger still when they came from different datasets. Because these protocols changed several factors together, the reductions cannot be attributed to task composition or dataset alone. Thus, for these operationally defined high- versus low-attention contrasts, technical compatibility and a common binary label space were insufficient to preserve target-specific performance under increasingly heterogeneous zero-shot transfer conditions.

The overall degradation pattern was observed across the five decoders, but the absolute level of balanced accuracy varied materially with decoder choice. This finding makes the decoder an important factor in comparative EEG evaluation rather than an interchangeable implementation detail. In particular, conclusions about a validation scenario should distinguish a pattern that is reproduced across model families from the performance of any single predeclared model recipe.

These results provide leakage-resistant, pre-adaptation baselines for assessing source–target suitability in the studied data compositions. They indicate that nominally aligned labels and partially harmonized acquisition specifications do not by themselves establish that a source dataset will support useful zero-shot transfer to a

target dataset or task composition. Such baselines can therefore inform source selection, which should be based on empirical, directed source–target comparisons rather than on nominal label alignment alone. By relating each transfer result to the corresponding target-specific baseline, the analysis distinguishes preservation of the contrast under transfer from the discriminability of that contrast in the target data itself. The resulting references also make it possible to evaluate whether subsequent transfer-learning, domain-adaptation, or personalization methods provide a meaningful improvement over matched zero-shot performance rather than merely achieving a favorable absolute score.

Several limitations delimit the interpretation of these findings. First, the binary labels represent protocol-defined contrasts and do not isolate attention from related differences in arousal, visual input, stress, or task demands. Second, selection of distinguish protocol conditions whose EEG differences may reflect task demands, stress, arousal, sensory input, or motor responses as well as attention. In SAM-40, active cognitive tasks are contrasted with rest while listening to music, so successful classification does not identify which of these factors drives discrimination. Second, the mental attention-state dataset contains only five participants, limiting population generalization and the precision of participant-level bootstrap intervals despite the many windows per participant. The use of only two technically comparable datasets enabled a common input space for cross-dataset evaluation, but limited the volume and diversity of data available for model fittingfurther limits the diversity of the evaluated conditions. The analysis also used the 12 shared channels, a fixed parsimonious preparation pipeline, and predeclared model recipes without hyperparameter or architecture selection to provide comparable pre-adaptation benchmarks rather than to maximize performance in each scenario. The reported figures consequently do not represent the maximum attainable performance for an individual scenario.

Future work could usefully optimize hyperparameters and architectural choices within training folds, which would help distinguish limitations of the predeclared recipes from those imposed by the data compositions and transfer scenarios. It would also be informative to assess whether domain-adaptation methods offer value relative to the same models without adaptation. Such comparisons could provide a clearer indication of the extent to which adaptation improves transfer-learning performance beyond the pre-adaptation benchmarks reported here. To preserve the interpretability of these estimates, future comparisons would benefit from retaining group-disjoint, leakage-resistant splits and from avoiding the use of target test data for model selection or parameter tuning.

References

1. Rashid M, Sulaiman N, Abdul Majeed APP, Musa RM, Ab Nasir AF, Bari BS, et al. Current status, challenges, and possible solutions of EEG-based brain–computer interface: A comprehensive review. Frontiers in Neurorobotics. 2020;14:25. doi:10.3389/fnbot.2020.00025.

2. Wu D, Xu Y, Lu BL. Transfer learning for EEG-based brain–computer interfaces: A review of progress made since 2016. IEEE Transactions on Cognitive and Developmental Systems. 2022;14(1):4-19. doi:10.1109/TCDS.2020.3007453.

3. Siino M, Bonomo G, Sorbello R, Tinnirello I. Investigating the Impact of Rational Dilated Wavelet Transform on Motor Imagery EEG Decoding with Deep Learning Models. IEEE Access. 2025;13:214223-35. doi:10.1109/ACCESS.2025.3645762.

4. Sun Q, Zhou Y, Gong P, Zhang D. Attention detection using EEG signals and machine learning: A review. Machine Intelligence Research. 2025;22(2):219-38. doi:10.1007/s11633-024-1492-6.

5. Li H, Chen Y, Wang Y, Ni W, Zhang H. Foundation models for cross-domain EEG analysis application: A survey. arXiv preprint arXiv:250815716. 2025. arXiv:2508.15716. doi:10.48550/arXiv.2508.15716.

6. Zhang K, Xu G, Zheng X, Li H, Zhang S, Yu Y, et al. Application of transfer learning in EEG decoding based on brain–computer interfaces: A review. Sensors. 2020;20(21):6321. doi:10.3390/s20216321.

7. Hommel B, Chapman CS, Cisek P, Neyedli HF, Song JH, Welsh TN. No one knows what attention is. Attention, Perception, & Psychophysics. 2019;81(7):2288-303. doi:10.3758/s13414-019-01846-w.

8. Knudsen EI. Fundamental Components of Attention. Annual Review of Neuroscience. 2007;30:57-78. doi:10.1146/annurev.neuro.30.051606.094256.

9. Petersen SE, Posner MI. The Attention System of the Human Brain: 20 Years After. Annual Review of Neuroscience. 2012;35:73-89. doi:10.1146/annurev-neuro-062111-150525.

10. Zhou Y, Xu Z, Niu Y, Wang P, Wen X, Wu X, et al. Cross-Task Cognitive Workload Recognition Based on EEG and Domain Adaptation. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2022;30:50-60. doi:10.1109/TNSRE.2022.3140456.

11. Song M, Gwon D, Jun SC, Ahn M. Signal alignment for cross-datasets in P300 brain–computer interfaces. Journal of Neural Engineering. 2024;21(3):036007. doi:10.1088/1741-2552/ad430d.

12. Siino M, Bonomo G, Sorbello R, Tinnirello I. RatioWaveNet: A Learnable RDWT Front-End for Robust and Interpretable EEG Motor-Imagery Classification; 2025. Available from: `https://arxiv.org/abs/2510.21841`. arXiv:2510.21841. doi:10.48550/arXiv.2510.21841.

13. Zhang W, Deng L, Zhang L, Wu D. A Survey on Negative Transfer. IEEE/CAA Journal of Automatica Sinica. 2023;10(2):305-29. doi:10.1109/JAS.2022.106004.

14. Acı Çİ, Kaya M, Mishchenko Y. Distinguishing mental attention states of humans via an EEG-based passive BCI using machine learning methods. Expert Systems with Applications. 2019;134:153-66. doi:10.1016/j.eswa.2019.05.057.

15. Ghosh R, Deb N, Sengupta K, Phukan A, Choudhury N, Kashyap S, et al. SAM 40: Dataset of 40 subject EEG recordings to monitor the induced-stress while performing Stroop color-word test, arithmetic task, and mirror image recognition task. Data in Brief. 2022;40:107772. doi:10.1016/j.dib.2021.107772.

16. Chen X, Bao X, Jitian K, Li R, Zhu L, Kong W. Hybrid EEG Feature Learning Method for Cross-Session Human Mental Attention State Classification. Brain Sciences. 2025;15(8):805. doi:10.3390/brainsci15080805.

17. Lan Z, Sourina O, Wang L, Scherer R, Müller-Putz GR. Domain adaptation techniques for EEG-based emotion recognition: A comparative study on two public datasets. IEEE Transactions on Cognitive and Developmental Systems. 2019;11(1):85-94. doi:10.1109/TCDS.2018.2826840.

18. MacLeod CM. Half a century of research on the Stroop effect: An integrative review. Psychological Bulletin. 1991;109(2):163-203. doi:10.1037/0033-2909.109.2.163.

19. Scarpina F, Tagini S. The Stroop color and word test. Frontiers in Psychology. 2017;8:557. doi:10.3389/fpsyg.2017.00557.

20. Esterman M, Noonan SK, Rosenberg M, DeGutis J. In the zone or zoning out? Tracking behavioral and neural fluctuations during sustained attention. Cerebral Cortex. 2013;23(11):2712-23. doi:10.1093/cercor/bhs261.

21. Grabner RH, Ansari D, Reishofer G, Stern E, Ebner F, Neuper C. Individual differences in mathematical competence predict parietal brain activation during mental calculation. NeuroImage. 2007;38(2):346-56. doi:10.1016/j.neuroimage.2007.07.041.

22. Kitaura Y, Nishida K, Yoshimura M, Mii H, Katsura K, Ueda S, et al. Functional localization and effective connectivity of cortical theta and alpha oscillatory activity during an attention task. Clinical Neurophysiology Practice. 2017;2:193-200. doi:10.1016/j.cnp.2017.09.002.

23. Shepard RN, Metzler J. Mental rotation of three-dimensional objects. Science. 1971;171(3972):701-3. doi:10.1126/science.171.3972.701.

24. Hyun JS, Luck SJ. Visual working memory as the substrate for mental rotation. Psychonomic Bulletin & Review. 2007;14(1):154-8. doi:10.3758/BF03194043.

25. Tomasino B, Gremese M. Effects of stimulus type and strategy on mental rotation network: An activation likelihood estimation meta-analysis. Frontiers in Human Neuroscience. 2016;9:693. doi:10.3389/fnhum.2015.00693.

26. Kamrud A, Borghetti B, Schubert Kabban C, Miller M. Generalized deep learning EEG models for cross-participant and cross-task detection of the vigilance decrement in sustained attention tasks. Sensors. 2021;21(16):5617. doi:10.3390/s21165617.

27. Ke Y, Jiang T, Liu S, Cao Y, Jiao X, Jiang J, et al. Cross-task consistency of electroencephalography-based mental workload indicators: Comparisons between power spectral density and task-irrelevant auditory event-related potentials. Frontiers in Neuroscience. 2021;15:703139. doi:10.3389/fnins.2021.703139.

28. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 785-94. doi:10.1145/2939672.2939785.

29. Schirrmeister RT, Springenberg JT, Fiederer LDJ, Glasstetter M, Eggensperger K, Tangermann M, et al. Deep learning with convolutional neural networks for EEG decoding and visualization. Human Brain Mapping. 2017;38(11):5391-420. doi:10.1002/hbm.23730.

30. Lawhern VJ, Solon AJ, Waytowich NR, Gordon SM, Hung CP, Lance BJ. EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. Journal of Neural Engineering. 2018;15(5):056013. doi:10.1088/1741-2552/aace8c.

31. Song Y, Zheng Q, Liu B, Gao X. EEG Conformer: convolutional transformer for EEG decoding and visualization. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2023;31:710-9. doi:10.1109/TNSRE.2022.3230250.

Supporting information

S1 Table. Performance metrics across experimental scenarios for XGBoost.

S2 Table. Performance metrics across experimental scenarios for EEGNet.

S3 Table. Performance metrics across experimental scenarios for ShallowFBCSPNet.

S4 Table. Performance metrics across experimental scenarios for EEGConformer.

S1 Table. Performance metrics across experimental scenarios for XGBoost.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.67	0.35	0.74	0.64/0.70
	Stratified K-fold(Full B)	0.68	0.65	0.36	0.75	0.60/0.76
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.76	0.68/0.70
	Stratified K-fold(B: Arithmetic)	0.70	0.68	0.41	0.76	0.69/0.70
	Stratified K-fold(B: Mirror)	0.70	0.70	0.43	0.77	0.72/0.69
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.65	0.33	0.74	0.65/0.67
	Leave-one-subject-out(Full B)	0.65	0.61	0.31	0.73	0.58/0.72
Cross-task (Dataset B)	Full B → Stroop	0.64	0.61	0.31	0.71	0.60/0.68
	Full B → Arithmetic	0.65	0.63	0.33	0.71	0.59/0.71
	Full B → Mirror	0.67	0.65	0.37	0.77	0.58/0.77
	Stroop → Full B	0.61	0.54	0.22	0.67	0.63/0.58
	Stroop → Arithmetic	0.56	0.52	0.13	0.58	0.67/0.45
	Stroop → Mirror	0.62	0.59	0.27	0.69	0.65/0.59
	Arithmetic → Full B	0.64	0.56	0.25	0.70	0.68/0.59
	Arithmetic → Stroop	0.58	0.55	0.18	0.62	0.68/0.48
	Arithmetic → Mirror	0.67	0.64	0.36	0.74	0.66/0.67
	Mirror → Full B	0.65	0.58	0.28	0.73	0.71/0.59
	Mirror → Stroop	0.61	0.58	0.25	0.67	0.72/0.51
	Mirror → Arithmetic	0.64	0.62	0.32	0.72	0.70/0.59
Cross-dataset	Full A → Full B	0.52	0.49	0.04	0.52	0.28/0.75
	Full B → Full A	0.54	0.52	0.09	0.55	0.37/0.71
	Full A → Stroop	0.48	0.42	-0.04	0.47	0.26/0.70
	Stroop → Full A	0.51	0.48	0.02	0.50	0.32/0.69
	Full A → Arithmetic	0.54	0.47	0.10	0.55	0.27/0.81
	Arithmetic → Full A	0.59	0.58	0.18	0.62	0.63/0.55
	Full A → Mirror	0.51	0.45	0.05	0.52	0.27/0.76
	Mirror → Full A	0.53	0.52	0.07	0.56	0.61/0.45

S2 Table. Performance metrics across experimental scenarios for EEGNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.66	0.34	0.74	0.57/0.76
	Stratified K-fold(Full B)	0.67	0.64	0.33	0.74	0.58/0.76
	Stratified K-fold(B: Stroop)	0.66	0.64	0.35	0.73	0.59/0.73
	Stratified K-fold(B: Arithmetic)	0.65	0.64	0.33	0.71	0.63/0.68
	Stratified K-fold(B: Mirror)	0.67	0.66	0.36	0.74	0.62/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.65	0.63	0.32	0.76	0.62/0.68
	Leave-one-subject-out(Full B)	0.66	0.62	0.33	0.75	0.59/0.73
Cross-task (Dataset B)	Full B → Stroop	0.66	0.63	0.35	0.74	0.61/0.70
	Full B → Arithmetic	0.65	0.62	0.33	0.73	0.59/0.70
	Full B → Mirror	0.68	0.67	0.40	0.76	0.60/0.77
	Stroop → Full B	0.59	0.53	0.19	0.67	0.57/0.62
	Stroop → Arithmetic	0.54	0.50	0.08	0.59	0.62/0.45
	Stroop → Mirror	0.59	0.55	0.19	0.66	0.59/0.59
	Arithmetic → Full B	0.62	0.56	0.24	0.69	0.61/0.63
	Arithmetic → Stroop	0.59	0.55	0.20	0.64	0.62/0.56
	Arithmetic → Mirror	0.64	0.62	0.32	0.70	0.63/0.66
	Mirror → Full B	0.64	0.60	0.27	0.69	0.62/0.66
	Mirror → Stroop	0.61	0.59	0.24	0.65	0.61/0.61
	Mirror → Arithmetic	0.63	0.61	0.30	0.68	0.61/0.66
Cross-dataset	Full A → Full B	0.47	0.34	-0.07	0.46	0.61/0.32
	Full B → Full A	0.51	0.48	0.04	0.53	0.30/0.72
	Full A → Stroop	0.45	0.37	-0.13	0.39	0.60/0.29
	Stroop → Full A	0.50	0.46	0.01	0.50	0.27/0.74
	Full A → Arithmetic	0.49	0.43	-0.00	0.52	0.58/0.40
	Arithmetic → Full A	0.50	0.43	0.01	0.51	0.16/0.84
	Full A → Mirror	0.46	0.38	-0.10	0.46	0.60/0.32
	Mirror → Full A	0.49	0.48	-0.02	0.48	0.44/0.53

S3 Table. Performance metrics across experimental scenarios for ShallowFBCSPNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.66	0.34	0.74	0.64/0.68
	Stratified K-fold(Full B)	0.72	0.72	0.46	0.82	0.57/0.87
	Stratified K-fold(B: Stroop)	0.78	0.78	0.59	0.86	0.77/0.80
	Stratified K-fold(B: Arithmetic)	0.76	0.76	0.54	0.84	0.76/0.76
	Stratified K-fold(B: Mirror)	0.74	0.73	0.49	0.81	0.74/0.74
Cross-subject	Leave-one-subject-out(Full A)	0.67	0.66	0.36	0.77	0.71/0.63
	Leave-one-subject-out(Full B)	0.67	0.64	0.38	0.81	0.54/0.80
Cross-task (Dataset B)	Full B → Stroop	0.67	0.63	0.38	0.80	0.55/0.79
	Full B → Arithmetic	0.68	0.64	0.40	0.80	0.54/0.81
	Full B → Mirror	0.67	0.64	0.37	0.80	0.56/0.78
	Stroop → Full B	0.65	0.59	0.29	0.75	0.68/0.63
	Stroop → Arithmetic	0.59	0.55	0.21	0.67	0.73/0.45
	Stroop → Mirror	0.64	0.61	0.31	0.72	0.74/0.54
	Arithmetic → Full B	0.68	0.61	0.35	0.79	0.74/0.62
	Arithmetic → Stroop	0.63	0.59	0.28	0.72	0.74/0.52
	Arithmetic → Mirror	0.71	0.69	0.45	0.80	0.76/0.66
	Mirror → Full B	0.68	0.62	0.33	0.78	0.69/0.66
	Mirror → Stroop	0.63	0.61	0.28	0.70	0.70/0.57
	Mirror → Arithmetic	0.67	0.65	0.37	0.77	0.67/0.68
Cross-dataset	Full A → Full B	0.49	0.28	-0.05	0.49	0.83/0.16
	Full B → Full A	0.55	0.48	0.11	0.58	0.28/0.81
	Full A → Stroop	0.47	0.36	-0.08	0.41	0.82/0.13
	Stroop → Full A	0.56	0.52	0.12	0.60	0.52/0.59
	Full A → Arithmetic	0.52	0.42	0.04	0.57	0.79/0.24
	Arithmetic → Full A	0.56	0.53	0.14	0.61	0.55/0.56
	Full A → Mirror	0.48	0.37	-0.08	0.49	0.82/0.13
	Mirror → Full A	0.55	0.53	0.11	0.58	0.53/0.57

S4 Table. Performance metrics across experimental scenarios for EEGConformer.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.64	0.29	0.73	0.61/0.67
	Stratified K-fold(Full B)	0.72	0.71	0.44	0.81	0.60/0.84
	Stratified K-fold(B: Stroop)	0.73	0.72	0.48	0.81	0.64/0.82
	Stratified K-fold(B: Arithmetic)	0.71	0.69	0.46	0.81	0.58/0.84
	Stratified K-fold(B: Mirror)	0.69	0.67	0.41	0.79	0.57/0.81
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.65	0.33	0.75	0.65/0.66
	Leave-one-subject-out(Full B)	0.71	0.68	0.42	0.82	0.61/0.81
Cross-task (Dataset B)	Full B → Stroop	0.71	0.68	0.44	0.80	0.61/0.80
	Full B → Arithmetic	0.70	0.68	0.44	0.81	0.60/0.80
	Full B → Mirror	0.73	0.71	0.48	0.82	0.63/0.82
	Stroop → Full B	0.62	0.56	0.24	0.72	0.59/0.64
	Stroop → Arithmetic	0.56	0.51	0.12	0.61	0.66/0.45
	Stroop → Mirror	0.61	0.58	0.25	0.71	0.62/0.60
	Arithmetic → Full B	0.67	0.63	0.33	0.77	0.64/0.71
	Arithmetic → Stroop	0.62	0.59	0.26	0.69	0.61/0.62
	Arithmetic → Mirror	0.69	0.67	0.41	0.79	0.63/0.76
	Mirror → Full B	0.65	0.60	0.28	0.75	0.59/0.71
	Mirror → Stroop	0.61	0.58	0.23	0.68	0.58/0.63
	Mirror → Arithmetic	0.65	0.61	0.32	0.75	0.57/0.72
Cross-dataset	Full A → Full B	0.47	0.31	-0.07	0.43	0.70/0.24
	Full B → Full A	0.58	0.56	0.18	0.64	0.62/0.54
	Full A → Stroop	0.42	0.34	-0.18	0.33	0.65/0.20
	Stroop → Full A	0.52	0.43	0.07	0.62	0.13/0.91
	Full A → Arithmetic	0.53	0.45	0.05	0.56	0.60/0.45
	Arithmetic → Full A	0.59	0.57	0.20	0.66	0.68/0.50
	Full A → Mirror	0.47	0.39	-0.05	0.43	0.66/0.29
	Mirror → Full A	0.54	0.50	0.08	0.57	0.76/0.31